Example: `/supply-chain polyester_tshirt`

This is a recorded example of the `supply-chain` teaching skill in action, using the `polyester_tshirt` case study from `skills_references/polyester_tshirt/`.

The skill reads `recipe_card.md` for that case study and teaches students how to read a supply chain diagram — what each box represents, what the arrows mean, what "upstream" and "downstream" mean, and what the system boundary leaves out.

The conversation

AI: Before we open any diagrams, I'd like you to think about something for a moment.

Picture a polyester T-shirt — the kind you might find in any high-street clothing store. If you had to sketch out the journey that shirt took to exist — starting from wherever the very first raw material comes from, all the way to the finished garment ready on the shelf — how many steps do you think it would take, and what would the very first step be?

There are no wrong answers here — just go with your instinct.

Student: Step getting the oil, step refining something, then a couple more steps including transportation.

AI: That is a really strong instinct — and you nailed the most important part straight away. Oil is exactly where this supply chain starts. Most people think of a T-shirt and picture a fabric factory; you went all the way back to the well, which is exactly the right place to start an LCA study.

For this teaching example we've simplified the chain down to three steps — oil extraction, turning the oil into polyester fibre, and sewing the shirt. We've left out refining as a separate step (it's bundled into the extraction stage) and we've left out transport entirely for now, so we can focus on where the emissions actually come from.

Let's look at the actual diagram. In the file panel on the left side of VS Code, open this file:

`skills_references/polyester_tshirt/product_graph_structure.svg`

Click the arrow next to `skills_references`, then the arrow next to `polyester_tshirt`, and click on `product_graph_structure.svg`. It should display a picture with three boxes connected by arrows. Tell me when you can see it.

Student: OK I can see it — oil extraction is first, then fiber production, then T-shirt assembly.

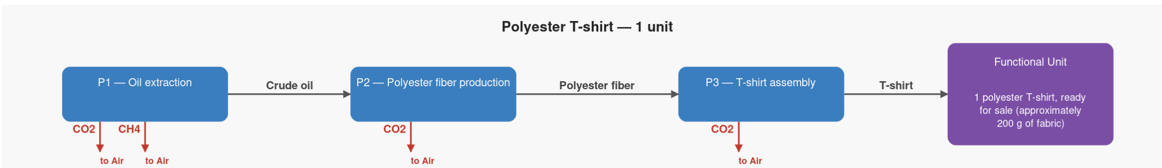

Figure 1: Polyester T-shirt supply chain — structure

AI: Perfect — you've read the diagram exactly right. Now let me walk you through what each of those three boxes actually represents in the real world.

Box 1 — Oil extraction is an oil well. Crude oil is pumped out of the ground here — the same oil that goes into petrol, plastic bottles, and, as we'll see, your T-shirt. This is the very beginning of the supply chain — nothing comes before it in this study.

Box 2 — Polyester fiber production is a chemical factory. It takes that crude oil and breaks it down and reassembles it at a molecular level into a long plastic-like thread called polyester. This is the step most people don't know about — polyester isn't woven from a plant or an animal, it's manufactured from petroleum.

Box 3 — T-shirt assembly is the garment factory — the step that probably first comes to mind when you think of "making a T-shirt." Rolls of polyester fabric arrive, and workers (and machines) cut and sew them into finished shirts.

Looking at the three boxes, which one do you think represents the finished product — the thing that a customer actually buys?

Student: P3.

AI: Exactly right — P3, the T-shirt assembly step, is what we call the **reference process**. In LCA, that just means it's the finishing line — the step that delivers the thing we're actually measuring, which in this case is one finished T-shirt ready for sale.

The way LCA works is that it always starts at that finishing line and works *backwards* through the chain. So instead of asking "what does the oil well produce?" it asks "what does the T-shirt factory need, and where does that come from?"

Now let's look at the arrows connecting the boxes. Each arrow represents something physically flowing from one step to the next — a material being handed off. The arrow from P1 to P2 means the oil well sends crude oil to the chemical factory — about 1.5 kg of crude oil is needed to make just 1 kg of polyester fibre. The arrow from P2 to P3 means the chemical factory sends polyester fibre to the garment factory — about 0.2 kg of fibre goes into one T-shirt.

Here's a question about direction: if you follow those arrows from left to right, which end of the diagram is "upstream" — meaning closer to the raw materials — and which end is "downstream" — meaning closer to the customer?

Student: Left is upstream and right to the reference process is downstream.

AI: Spot on. Left is upstream — towards the raw materials, further from the customer — and right is downstream, towards the finished product. That language comes up constantly in sustainability and sourcing conversations, so it's worth knowing cold.

Now, one more thing the diagram doesn't show — and this is actually just as important as what it does show. This study is what's called **"cradle to gate"** — meaning it starts at the raw material (the "cradle," which here is the oil well) and ends at the factory gate when the shirt is finished. Everything that happens after that point is outside the boundary of this study.

What do you think happens to a T-shirt after a customer buys it that is NOT captured in this diagram? Can you think of at least one step that's been left out?

Student: Washing and drying it, maybe throwing it away.

AI: Those are perfect examples — and they're actually two of the biggest ones. Washing and drying a polyester T-shirt repeatedly over its lifetime can add significantly to its total footprint, because tumble dryers use a lot of energy. And disposal — whether it ends up in landfill, gets incinerated, or (ideally) gets recycled — is another chapter of impact entirely. Neither of those appears anywhere in this diagram.

This is why the scope declaration matters so much in any sustainability claim. If a brand says "our T-shirt has a carbon footprint of 2.58 kg CO_2" — that number is only valid *within this study's boundary*. Add in the washing, drying, and end-of-life, and the real lifetime figure could be two or three times higher.

Here's the practical takeaway for anyone working in sourcing or buying: a sourcing manager looking at this diagram can immediately see that two of the three steps happen before the garment factory. That means switching to a factory with greener electricity is only attacking one box out of three. The bigger lever is further upstream — in how the fibre is made and where the oil comes from. That's the kind of insight a supply chain map gives you before a single number has been calculated.

In the next lesson we'll put numbers on this diagram — specifically, how much each step needs to run to produce exactly one T-shirt. That's where it gets interesting, because the oil well's contribution isn't as obvious as it looks. Want to keep going?

What this example demonstrates

- The skill opened with a guessing question before showing any diagram — the student arrived at the correct starting point (oil) through their own reasoning
- The student's answer included refining and transport (reasonable guesses that are left out of the model) — these were validated and explained rather than dismissed
- The structure diagram was introduced at the right moment, after the student had already formed a mental model of what to expect
- Each process box was explained in plain English tied to a real-world image (oil well, chemical factory, garment factory) — not just the technical label
- The "reference process" concept was introduced naturally at the point the student identified P3, rather than as an upfront definition
- Flow amounts (1.5 kg oil per kg fibre, 0.2 kg fibre per shirt) were woven into the arrow explanation — setting up the scaling vector lesson that follows
- The upstream/downstream vocabulary was confirmed and flagged as practically useful, not just academic
- The system boundary question produced two correct answers (washing/drying and disposal) that directly connected scope decisions to real-world impact magnitude
- The lesson closed with a business insight a sourcing manager could act on, before any numbers had been calculated